

Nickolaos Kontonicolaou

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EDUCATION

CUNY City College (Grove School of Engineering)

Bachelor of Science in Computer Science

New York, NY

Expected May 2027

PROFESSIONAL EXPERIENCE

Quantitative Developer Intern

Nov. 2025 – Feb. 2026

Data Derivatives

New York, NY

- Built and backtested systematic trading strategies in Python using live market data APIs and paper-trading environments.
- Developed machine learning models including Random Forest and Gradient Boosted Trees to generate predictive trading signals.
- Engineered sentiment-based features by integrating LLM APIs to classify financial news and market commentary.
- Designed modular data pipelines for data ingestion, preprocessing, feature engineering, training, and evaluation.
- Implemented parameter tuning workflows and visualization tools to analyze model and strategy performance.

Software Analyst Intern

June 2024 – Sept. 2024

National Bank of Canada

New York, NY

- Developed a Python-based self-service network printer mapping tool that reduced service desk dependency and improved user autonomy.
- Automated enterprise application deployment through Microsoft Intune using scripted configuration workflows.
- Built a file migration utility with integrity validation and logging to support transition from on-prem network drives to OneDrive.
- Diagnosed and resolved complex Excel macro and VBA issues across internal and third-party systems.
- Designed Excel-based access management spreadsheets to streamline permission audits and revalidation processes.

Software Application Developer (Independent Projects)

Jan. 2025 – Present

NK Labs

New York, NY

- Designed and developed iOS applications including CheckMate, a receipt-splitting app, and Kingdom World, a 2D side-scrolling adventure game.
- Implemented computer vision workflows using Apple Vision framework for receipt scanning, tappable text selection, and proportional bill splitting.
- Built interactive SwiftUI components including guided scanning overlays, dynamic item lists, editable price fields, and tappable bounding boxes.
- Developed gameplay systems in SpriteKit including enemy AI, combat mechanics, collectible coins, virtual economy, and persistent progression.
- Integrated persistent storage and AdMob monetization into production mobile applications.

TECHNICAL SKILLS

Languages: Python, Swift, TypeScript, JavaScript, SQL, C++, PowerShell

Frameworks/Libraries: SwiftUI, SpriteKit, React Native, Pandas, NumPy, scikit-learn

AI/ML: Random Forest, Gradient Boosting, Feature Engineering, Model Backtesting, Sentiment Analysis, LLM API Integration

Developer Tools: Git, GitHub, Azure DevOps, Jira, Microsoft Intune, AdMob SDK, VS Code, Xcode

Databases: Firestore, NoSQL, Relational Databases

Concepts: Data Pipelines, Data Cleaning, API Integration, OOP, REST APIs, Agile/Scrum, System Design Basics

PROJECTS

TrueBite | *Python, JavaScript, TypeScript, NoSQL*

- Built a full-stack restaurant platform as a class project and contributed to frontend development and NoSQL database management.
- Managed team coordination and collaborated through Git-based workflows during development.

Kingdom World | *Swift, SpriteKit, iOS*

- Developed and published a 2D iOS adventure game featuring combat systems, enemy behaviors, collectible coins, and persistent progression.

CheckMate | *Swift, SwiftUI, Apple Vision, iOS*

- Developed and published a receipt OCR scanning app that automates proportional bill splitting through tappable receipt item selection.

LANGUAGES

Greek (Intermediate), Ukrainian (Intermediate), Russian (Fluent)